

Out-Of-Core Diffraction for Terascale Holography

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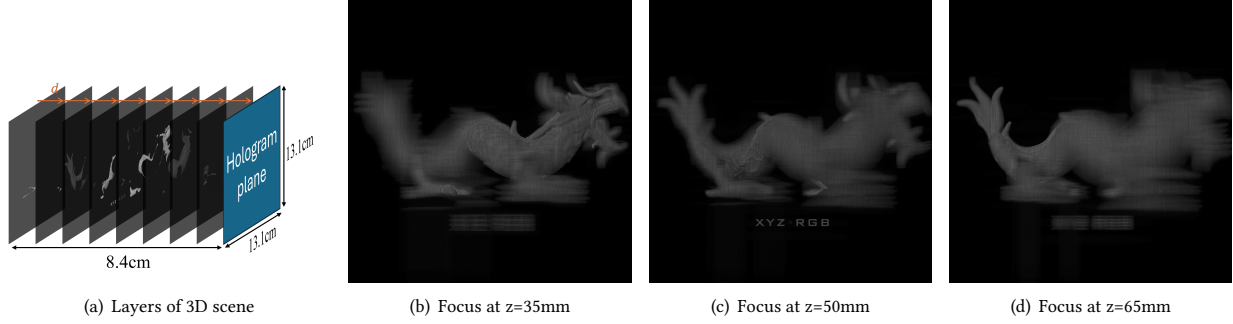


Figure 1: The resolution of the hologram is $256K^2$ (physical volume: $13.1 \times 13.1 \times 8.4 \text{ cm}^3$), and it requires up to 4 TB of memory space for the CGH process. Our method generates it with 64 GB system memory and 24 GB GPU memory, achieving up to 4.2 times higher performance than prior methods. (b)-(c) show the numerically reconstructed results with different focal distances.

Abstract

Displaying large-scale holograms with a wide field of view (FoV) requires ultra-high-resolution data, often reaching tera-scale sizes. We propose an out-of-core diffraction method that utilizes multiple SSDs simultaneously to manage tera-scale holography within limited memory constraints. To enhance I/O bandwidth, we introduce a block-wise I/O approach for diffraction computation. Additionally, our method employs a compressed data transfer scheme to reduce I/O costs. As a result, our method performs diffraction up to 4.2 faster compared to alternative methods. We also validate our approach by generating a $256K^2$ resolution hologram, which would typically require 4TB of memory using conventional diffraction methods, on a GPU with only 24 GB of memory.

CCS Concepts

• **Computing methodologies** → *Rendering; Massively parallel and high-performance simulations; Scientific visualization.*

Keywords

computer-generated holography, out-of-core, external-memory, diffraction, CGH

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1 Introduction

Holography is considered one of the next-generation display technologies because it can record and reproduce three-dimensional light wavefields through a two-dimensional medium, known as a hologram. Computer-generated holography (CGH) creates holograms of virtual objects by simulating physical light waves.

One of the major challenges in CGH is the generation of tera-scale holograms with resolutions reaching hundreds of thousands of pixels. These high resolutions are necessary to achieve sufficient display sizes with nano-scale pixels, enabling wide diffraction angles and a large FoV. However, the limited memory space of computing units poses a significant barrier to performing holographic wavefield simulations. In particular, calculating diffraction demands four times the memory space of the input matrix due to the use of zero-padding, which prevents circular convolution aliasing in FFT-based convolution.

One approach to addressing this issue is out-of-core (OOC) computation, which utilizes storage devices (e.g., SSDs) as supplementary memory space [Blinder and Shimobaba 2019]. However, the effectiveness of previous out-of-core diffraction methods has been constrained by I/O bottlenecks, even when data is distributed across multiple SSDs [Lee and Kim 2023].

In this work, we introduce a novel OOC diffraction algorithm designed to enhance I/O efficiency through a block-wise I/O approach and minimize I/O costs using a compressed data transfer scheme.

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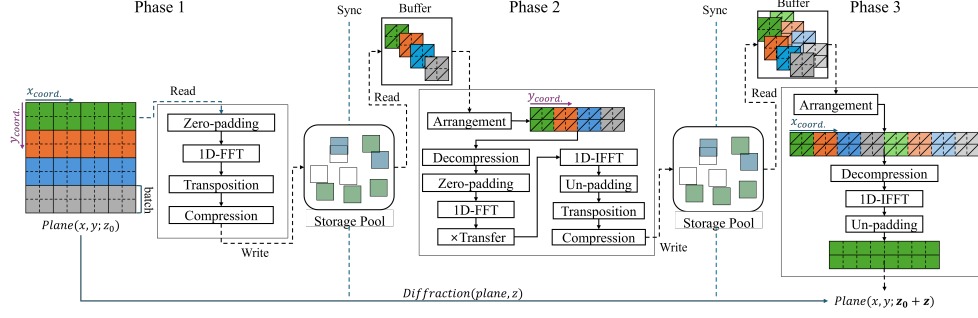


Figure 2: Processes overview of the proposed out-of-core diffraction method.

Our method enables the generation of a $256K^2$ hologram, which would typically require up to 4 TB of memory space, while delivering up to 4.2 times greater performance compared to previous methods (Fig. 1).

2 Our approach

Fig. 2 shows an overview of the proposed method. The input plane is stored on a storage device, as it is too large to be loaded into memory, with all computations performed on a GPU. The process consists of three phases, with write and read operations required for intermediate results between each phase. The major challenge in achieving efficient diffraction calculations for such tera-scale problems is minimizing the I/O bottleneck between the storage device and the GPU.

The 1D-FFT-based diffraction processes the plane strip-by-strip. To improve transaction efficiency, we decompose a strip of the plane into blocks and access it in a block-wise manner, enabling sequential data access on storage devices. Additionally, we reduce I/O costs by compressing the blocks using Zfp [Lindstrom 2014] and distributing them into a storage pool capable of utilizing up to hundreds of storage devices. As a result, our approach not only improves the bandwidth efficiency of SSDs but also reduces I/O costs based on the compression ratio and the number of storage devices. To further enhance performance, we pipeline independent read-compute-write operations, maximizing parallelism.

3 Results

To evaluate our method (*Ours*), we compared it with a strip-wise OOC (*SW-OOC*) diffraction method [Blinder and Shimobaba 2019] and an implicit convolution method [Lee and Kim 2023]. The test system included an Nvidia RTX 3090 GPU (24 GB), and six Samsung 980 Pro 2TB SSDs. The system memory functions primarily as a buffer between the SSDs and the GPU, utilizing up to 64 GB, including the operating system's usage. Depending on each algorithm's SSD scalability, we used two, four, and six SSDs for *SW-OOC*, implicit convolution, and *Ours*, respectively.

Fig. 3 shows the processing time of each algorithm for generating a $128K^2$ hologram, whose size is 256 GB and 128 GB in double- and single-precision, respectively. It requires up to 1 TB and 512 GB of memory space for diffraction computation. For *SW-OOC* and *Ours*, we also compare the results with different compression ratios.

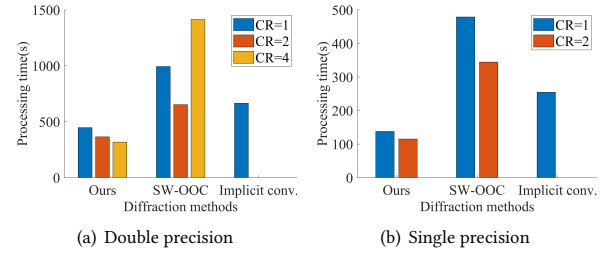


Figure 3: Comparison of the processing time for generating $128K^2$ holograms in single- and double-precision using three algorithms. CR denotes compression ratio.

As a result, *Ours* achieves up to 3.1x and 4.2x higher performance than *SW-OOC* in double- and single-precision, respectively. Compared to implicit convolution, *Ours* achieves approximately twice the performance for both precisions. With compression, our method continuously reduces processing time by up to 29% with a compression ratio of 4 compared to non-compressed cases.

To validate the scalability of our method, we generated a tera-scale hologram (i.e., $256K^2$) requiring 4 TB of memory for computation using layer-based CGH [Zhao et al. 2015]. Fig. 1 shows the layers of the 3D object and its numerically reconstructed image. It took only 18 hours using just 24 GB of GPU memory, compared to several days required by previous methods.

Acknowledgments

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